

CAF Core — Top-Performer Mimic Flows

(Complete Guide)

Version: CAF Core repository snapshot (June 2026)

Audience: Operators, engineers, and AI assistants (e.g. ChatGPT project folders)

Scope: `FLOW_TOP_PERFORMER_MIMIC_IMAGE` , `FLOW_TOP_PERFORMER_MIMIC_CAROUSEL` , and their upstream/downstream dependencies

1. Executive summary

Mimic flows are optional CAF pipeline lanes that recreate the **visual pattern** of archived **top-performer** social posts while generating **fresh copy** aligned to the brand and the current signal-pack idea. They are **not** pixel-perfect clones: logos, faces, and copyrighted imagery are explicitly excluded (`twist_brief.visual_only`).

The system splits work into two phases:

1. **Generate (copy)** — OpenAI LLM writes caption, hashtags, and slide text faithful to the reference **structure** (hook device, pacing, CTA shape).
2. **Render (visual)** — Image models (BFL FLUX, DashScope Qwen, NVIDIA NIM, or OpenAI `gpt-image-1`) produce plates; Puppeteer Handlebars templates or Sharp compositing overlay the LLM copy with typography derived from upstream vision analysis (Nemotron / Document AI).

Mimic flows sit in the standard CAF funnel:

```
Signal Pack → Candidates → Decision Engine → Content Job → Draft → Render → Review → Publish
```

Each mimic job is keyed by `task_id` like any other content job. Render metadata lives on `generation_payload.mimic_v1` (source of truth for render). Carousel mimic jobs additionally snapshot `draft_package_snapshot` with `package_type: "mimic_carousel_package"` — this is **distinct** from normal `FLOW_CAROUSEL` / `carousel_package`.

2. Flow types

Flow type	Platform format	Reference tier	When to use
<code>FLOW_TOP_PERFORMER_MIMIC_IMAGE</code>	Single post (post)	<code>top_performer_deep</code>	Idea grounded to a top performer with exactly one archived reference frame
<code>FLOW_TOP_PERFORMER_MIMIC_CAROUSEL</code>	Carousel	<code>top_performer_carousel</code>	Multi-slide decks, carousel-format ideas, or references with 2+ frames
<code>FLOW_TOP_PERFORMER_MIMIC_VIDEO</code>	(placeholder)	—	Not wired to render in current code

Routing rules:

- Image mimic **fails early** if more than one reference frame resolves.
- Carousel mimic handles multi-slide archives, video-slide skipping, and promotional-slide filtering.
- Mimic flows use **separate planning lanes** (`mimic_image` , `mimic_carousel`) so they can run **in parallel** with standard `FLOW_CAROUSEL` without sharing the `max_carousel_jobs_per_run` cap.

Code: `src/domain/top-performer-mimic-flow-types.ts` , `src/decision_engine/format-routing.ts`

3. Prerequisites (data + configuration)

3.1 Upstream data

1. **Top-performer ingest** — Creative Intelligence pipeline archives inspection media to Supabase (`stored_inspection_media_json` on `visual-guidelines` entries). See `docs/CREATIVE_INTELLIGENCE.md` .
2. **Signal pack** with `derived_globals_json.visual_guidelines_pack_v1` — per-insight rows carrying aesthetic analysis, slide transcripts, `mimic_evaluation` , and archived media paths.
3. **Ideas grounded** to top-performer `insights_id` values (`ideas_json[].grounding_insight_ids`). Each insight id may ground **at most one** idea.
4. For image mimic expansion at planning time: either explicit `target_flow_type` , `manual_mimic_pick` with `mimic_kind: "image"` , or eligibility via `mimicImageReferenceEligible()` (single-frame deep reference).

3.2 Environment flags

Variable	Default	Purpose			
MIMIC_IMAGE_ENABLED	false	Master switch — mimic draft/render paths stay off when false			
MIMIC_IMAGE_PROVIDER	bfl	bfl \	dashscope \	nvidia \	openai
MIMIC_IMAGE_BFL_MODEL	flux-2-klein-4b	BFL model slug (flux-2-flex for typography-tuned)			
MIMIC_VISUAL_SIMILARITY_PCT	70	How closely reference_edit should follow pixels (≤25% → low fidelity on OpenAI)			
MIMIC_IMAGE_INPUT_MODE	reference_edit	reference_edit (image-to-image) or analysis_t2i (text-to-image from Flux prompts)			
MIMIC_IMAGE_DEFAULT_SIZE	1024x1536	Instagram portrait default			
OPENAI_API_KEY	—	Required for copy generation (all mimic flows)			
Provider API keys	—	BFL_API_KEY , DASHSCOPE_API_KEY , NVIDIA_NIM_API_KEY , or OPENAI_API_KEY for render			

3.3 Project settings

- Enable FLOW_TOP_PERFORMER_MIMIC_IMAGE and/or FLOW_TOP_PERFORMER_MIMIC_CAROUSEL in allowed flow types.
 - Set plan cap > 0 for the mimic flow (flows are seeded **disabled** via seedMimicFlowTypesSkeleton()).
 - Optional per-project overrides in project_system_constraints : mimic_image_bfl_model , mimic_visual_similarity_pct , mimic_carousel_text_via_flux , mimic_image_input_mode .
- Migrations:** 060_mimic_flow_types.sql , 061_mimic_flow_prompts.sql , 062 – 069 (copy rules, grounding, BFL model, render settings).
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4. Core payload contracts

4.1 generation_payload.mimic_v1 (render source of truth)

Written during **mimic prep** (before or after LLM, depending on step). Consumed at **render**.

Field	Meaning		
mode	image_full \	template_bg \	carousel_visual
mode_override	Reviewer override from signal pack (mimic_mode_overrides)		
source_insights_id	Top-performer insight this job mimics		
reference_items[]	Archived frames: vision_fetch_url , bucket , object_path , source_slide_index		
archive_reference_items[]	Full deck before promo/video filtering		
slide_plans[]	Per output slide: render_mode (hbs \	full_bleed), reference_index	
visual_guideline	Slim vision row (no signed URLs)		
twist_brief	{ visual_only: true, legal_note }		
docai_layer_positions	Reviewer layout edits (preserved across re-renders)		
flux_image_prompts	Per-slide T2I prompts when image_input_mode = analysis_t2i		

Code: src/domain/mimic-payload.ts

4.2 mimic_carousel_package (review / operator snapshot)

Only FLOW_TOP_PERFORMER_MIMIC_CAROUSEL . Stored in draft_package_snapshot :

Slice	Content
copy	LLM slides, caption, hashtags
render_plan	strategy : template_background or per_slide_mimic
visual_reference	Storage paths, reference_items
visual_guideline	Format pattern, deck system, mimic_evaluation , per-slide cues
twist_brief	Legal / visual-only note

mimic_v1 remains authoritative for render; the package is for Review UI, content log, and operators.

Code: `src/domain/mimic-carousel-package.ts` , `src/services/draft-package-contract.ts`

4.3 Supporting payload keys

Key	When set	Purpose
<code>mimic_render_context</code>	Before LLM	Slide count targets, <code>copy_before_visual_mimic</code> , template path hints
<code>mimic_job_grounding</code>	Before LLM	<code>slide_copy_layout</code> — per-slide reference text, typography, copy slots
<code>template_storage_decision</code>	Before LLM	Whether backgrounds go to project template library
<code>template_backgrounds_prepared_at</code>	Pre-copy (template_bg)	Background plates extracted before copy gen
<code>candidate_data</code>	Planning	<code>grounding_insight_ids</code> , <code>idea_id</code> for lineage

5. Mimic modes and render strategies

5.1 Mode classification

`classifyMimicMode()` (`src/services/mimic-mode-classifier.ts`) picks the render path:

Priority:

1. Manual `mode_override` (reviewer on signal pack)
2. Nemotron `mimic_evaluation.recommended_mode` :
 - `text_on_template` → `template_bg`
 - `full_bleed_visual` → `carousel_visual`
3. Heuristic fallback (text density, `format_pattern` , unified background cues)

Mode	Typical reference	Render approach
<code>image_full</code>	1 frame	Single <code>STATIC_IMAGE</code> via image edit; on-image copy from LLM
<code>template_bg</code>	Text-heavy listicle / educational	Extract background plates → HBS (<code>carousel_mimic_bg.hbs</code>) or Sharp composite + text overlay
<code>carousel_visual</code>	Image-led / mixed deck	Per-slide art-only mimic (~70% similarity default) + HBS text overlay where <code>on_screen_text_transcript</code> exists

Slide render modes:

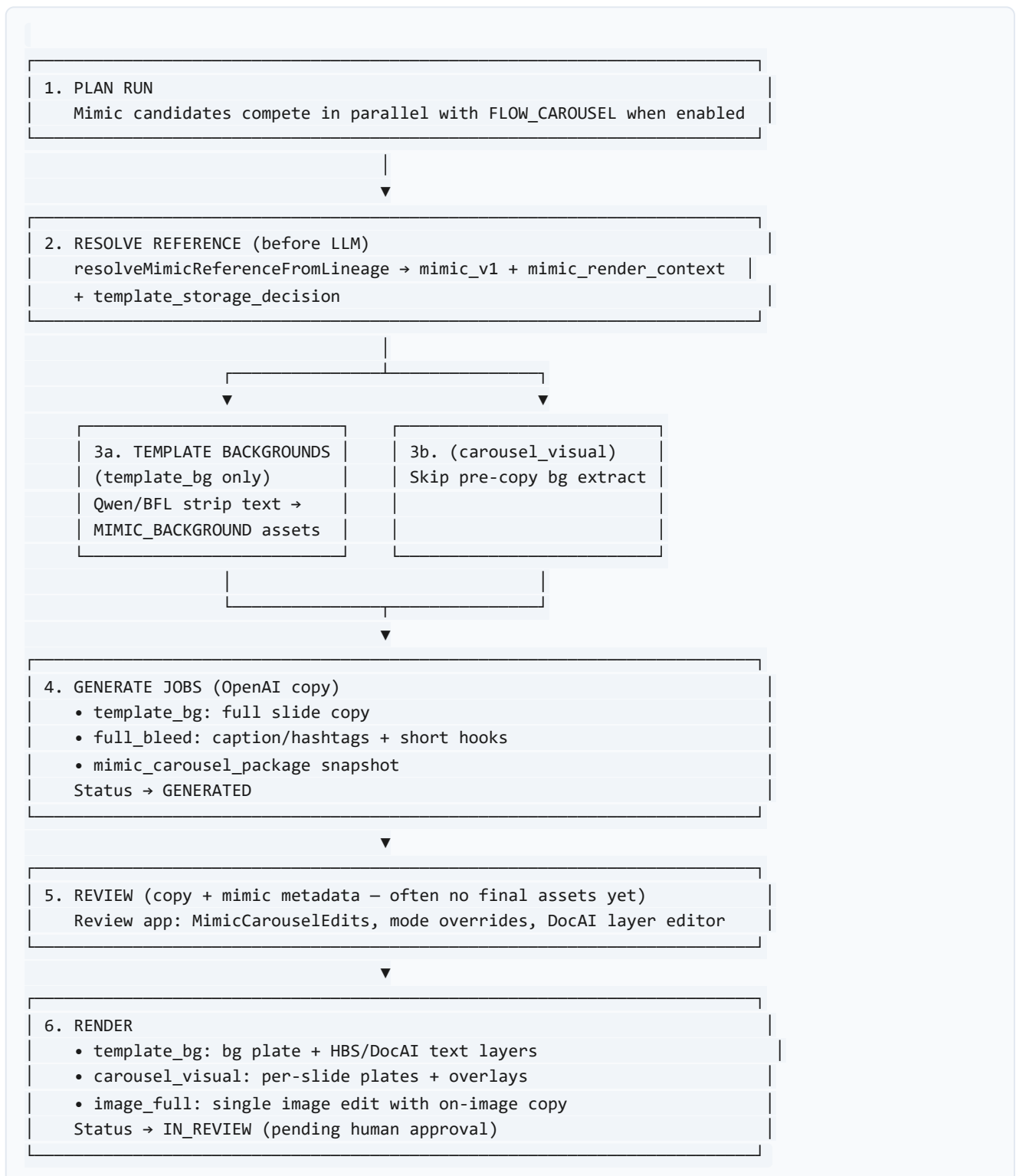
- `hbs` — Puppeteer Handlebars template overlays LLM copy; typography from Nemotron `text_blocks` + Document AI merge.
- `full_bleed` — Art-only image model output (no baked-in text); copy overlaid in a later HBS/DocAI step if needed.

Slides with **any** on-screen text transcript use **hbs**, not image-model typography.

5.2 Render plan strategies (carousel package)

render_plan.strategy	mimic_v1.mode	Behavior
template_background	template_bg	Reuse/extract bg plate → overlay text
per_slide_mimic	carousel_visual	Per-slide visual plate, then text overlay

6. End-to-end pipeline (operator workflow)



6.1 Key functions by stage

Stage	Primary modules
Reference resolve	<code>mimic-reference-resolver.ts</code> , <code>mimic-reference-eligibility.ts</code>
Pre-copy prep	<code>ensureMimicReferenceBeforeCopyGeneration</code> , <code>ensureMimicTemplateBackgroundsBeforeCopy</code> in <code>mimic-draft-prep.ts</code>
LLM copy	<code>llm-generator.ts</code> , <code>carousel-mimic-copy-policy.ts</code> , <code>mimic-copy-coherence.ts</code>
Post-copy snapshot	<code>prepareMimicDraftPackage</code> in <code>mimic-draft-prep.ts</code>
Carousel render	<code>mimic-carousel-render.ts</code> , <code>mimic-template-bg-render.ts</code> , <code>mimic-docai-overlay-layout.ts</code>
Image render	<code>mimic-image-job.ts</code> , <code>mimic-image-provider.ts</code>
Orchestration	<code>job-pipeline.ts</code>

7. Reference resolution (deep dive)

7.1 Lineage

Every mimic job links to a **signal pack** via `generation_payload` lineage (`getJobLineageByTaskId`). Resolution uses:

1. `candidate_data.grounding_insight_ids` (preferred), or
2. Job grounding rows from the signal pack.

7.2 Analysis tiers

Flow	Expected tier	Fallback
Image mimic	<code>top_performer_deep</code>	<code>top_performer_carousel</code> (flagged <code>reference_tier_fallback</code>)
Carousel mimic	<code>top_performer_carousel</code>	none

Image mimic **skips** entries with **>1** reference frame.

7.3 Reference items

Archived frames come from `stored_inspection_media_json` / `inspection_media` on the visual-guidelines entry. Each item carries:

- `vision_fetch_url` — fetch at render (re-signed from `bucket` + `object_path` when URLs expire)
- `source_slide_index` — 1-based position in original Instagram deck
- `is_video_slide` — skipped at render

`normalizeMimicReferenceItems()` enforces 1-based contiguous indexes for render plans.

7.4 Promotional / video slide handling

- **Video slides** — omitted from mimic reference set; indices tracked in `video_slide_indices`.
 - **Promotional slides** — `filterPromotionalSlidesFromMimicPayload()` drops slides tagged `self_promo`, `product_pitch`, or high `brand_specificity`.
 - **Skip indices** — `mimic_evaluation.skip_slide_indices` and `content_slide_indices` drive which slides count toward copy/render.
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8. LLM copy generation (deep dive)

8.1 Prompt templates

Migration `061_mimic_flow_prompts.sql` seeds:

- `MIMIC__Top_Performer_Carousel_v1` → `FLOW_TOP_PERFORMER_MIMIC_CAROUSEL`
- `MIMIC__Top_Performer_Image_v1` → `FLOW_TOP_PERFORMER_MIMIC_IMAGE`

Both **inherit the FLOW_CAROUSEL output schema** so QC and validation stay unchanged.

8.2 Signal pack budgeting (mimic-only)

Mimic prompts are large (vision grounding + top-performer knowledge). **Do not pass full** `ideas_json` (30+ rows). Required behavior:

1. Filter `ideas_json` to the job's planned idea only.
2. Match `idea_id` with underscore boundaries (avoid `idea_12` matching `idea_1`).
3. Slim `visual_guidelines_pack_v1` — strip `inspection_media` / signed URLs from LLM context.
4. Cap hashtag leaderboard size.

Code: `llm-creation-pack-budget.ts`, `llm-generator-helpers.ts` (`mimicFlowOnly: true`)

8.3 Copy branches

Mode	Copy expectation
<code>template_bg</code>	Full per-slide headline/body matching reference structure
<code>carousel_visual</code> / <code>full_bleed</code>	Caption, hashtags, short hooks; reference text length scaled by <code>MIMIC_FULL_BLEED_COPY_REFERENCE_SCALE</code>
<code>image_full</code>	Caption, hook, hashtags; on-image copy for render

8.4 Grounding blocks

`mimic_job_grounding.slide_copy_layout` provides per-slide:

- `reference_on_screen_text` (structure/length guide only — output must be fresh)
- `visual_description`, `layout_template`, `text_blocks`, `copy_slots_v1`

`assertMimicCopyDiffersFromReference()` guards against verbatim copying of reference wording.

Optional: `MIMIC_COPY_COHERENCE_LLM` runs a coherence pass after generation.

8.5 Context guards

Mimic carousel prompts have char budgets (`LLM_MIMIC_*_MAX_CHARS` , `MIMIC_CAROUSEL_SYSTEM_PROMPT_MAX_CHARS`) with emergency shrinking in `shrinkMimicCarouselPromptsIfNeeded()` .

9. Visual render (deep dive)

9.1 Image providers

`mimic-image-provider.ts` abstracts:

Provider	Typical model	Notes
bfl	<code>flux-2-klein-4b</code> , <code>flux-2-flex</code>	Default; async poll; flex supports steps/guidance
dashscope	<code>qwen-image-edit-max</code>	Alibaba; size <code>1024*1536</code>
nvidia	<code>qwen/qwen-image-edit</code>	NIM OpenAI-compatible <code>/images/edits</code>
openai	<code>gpt-image-1</code>	<code>input_fidelity</code> , <code>quality</code>

All prompts pass through `finalizeMimicImageModelPrompt()` with **art-only guard** (no on-image text unless explicitly allowed).

9.2 Template background path (`template_bg`)

1. **Pre-copy** (optional): `ensureMimicTemplateBackgroundsBeforeCopy` extracts cover/body/CTA plates → assets type `MIMIC_BACKGROUND` .
2. **Reusable templates**: when `mimic_evaluation.template_storage_quality = reusable` , plates also go to project template library (`mimic-template-library.ts`).
3. **Render**: Puppeteer renders `carousel_mimic_bg.hbs` with LLM copy + Nemotron typography, **or** Sharp composite (`carousel-composite-render.ts`) when `CAROUSEL_COMPOSITE_ENABLED=true` .

9.3 Carousel visual path (`carousel_visual`)

1. Per slide: `generateMimicSlideImage()` with art-only prompt (`mimic-prompt-builder.ts`).
2. Reference frame selected via `referenceItemForMimicSlide()` using `source_slide_index` alignment.
3. Output assets: `MIMIC_VISUAL_PLATE` per slide.
4. Text overlay: Document AI + Nemotron merge → HBS or DocAI fit page (`mimic-docai-fit-page.ts` , `mimic-docai-text-contrast.ts`).

9.4 Image full path (`image_full`)

`processImageMimicJob()` :

1. Asserts single reference frame.
2. Reads on-image copy from `generated_output` (`onImageCopyForMimicRender`).
3. One image edit → `STATIC_IMAGE` asset.

4. Job status → `IN_REVIEW`.

9.5 Re-render and text-only reprint

Reviewers can adjust `docai_layer_positions` and trigger **text-overlay reprint** (`MIMIC_TEXT_OVERLAY_REPRINT_PHASE`) without re-running image models — requires existing `MIMIC_BACKGROUND` or `MIMIC_VISUAL_PLATE` assets.

9.6 `analysis_t2i` mode

When `MIMIC_IMAGE_INPUT_MODE=analysis_t2i`:

- Per-slide Flux prompts generated at copy time (`mimic-flux-image-prompts.ts`, optional `MIMIC_FLUX_PROMPT_LLM`).
 - Render uses text-to-image (no reference pixels) — useful when reference edit is undesirable.
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10. Planning and candidate materialization

10.1 Image mimic expansion guard

`shouldExpandTopPerformerMimicImageForRow()` returns true when:

- `manual_mimic_pick === true` and `mimic_kind === "image"`, or
- `target_flow_type === FLOW_TOP_PERFORMER_MIMIC_IMAGE`, or
- Grounding resolves to single-frame deep reference (`mimicImageReferenceEligible`).

Otherwise image mimic candidates are **skipped** at expansion.

10.2 Parallel lanes

`format-routing.ts` assigns:

- `mimic_carousel` lane for carousel mimic
- `mimic_image` lane for image mimic
- Standard `carousel` lane for `FLOW_CAROUSEL`

This allows one idea to produce both a templated carousel job and a mimic carousel job when both flows are enabled.

10.3 Creative intelligence boost

Ideas with `grounding_insight_ids` starting with `ci_` get higher `past_performance` score (`CREATIVE_INTEL_PLANNER_PAST_PERFORMANCE_BOOST`).

11. Review app integration (`apps/review`)

Feature	API / component
Mimic carousel edits	<code>MimicCarouselEdits.tsx</code>
DocAI layer position editor	<code>MimicDocAiLayerPositionEditor.tsx</code>
Mode override per insight	<code>POST .../signal-packs/:packId/mimic-mode-override</code>
Mimic image audits	<code>GET .../mimic-image-audits</code>
Layer editor panel	<code>MimicCarouselLayerEditorPanel.tsx</code>

Review app is a **client** of Core APIs — `content_jobs` in Postgres remains source of truth.

12. Asset types

Asset type	When created
<code>MIMIC_BACKGROUND</code>	Template-bg plate (cover/body/CTA)
<code>MIMIC_VISUAL_PLATE</code>	Per-slide art-only plate (carousel_visual)
<code>STATIC_IMAGE</code>	Single-frame image mimic output
<code>CAROUSEL_IMAGE</code>	Final composed slide images

13. Configuration reference (extended)

Variable	Purpose
<code>MIMIC_CAROUSEL_TEXT_VIA_FLUX</code>	Legacy Flux text-bake (pipeline forces DocAI/HBS overlay today)
<code>MIMIC_USE_PROJECT_BRAND_PALETTE</code>	Inject brand colors into mimic prompts
<code>MIMIC_USE_BRAND_IMAGE_STYLE_HINTS</code>	Brand image style in prompts
<code>MIMIC_FLUX_PROMPT_LLM</code>	OpenAI authors Flux T2I prompts
<code>MIMIC_FULL_BLEED_COPY_REFERENCE_SCALE</code>	Scales reference text length targets
<code>MIMIC_COPY_CHAR_SLACK</code>	± chars tolerance on copy length guards
<code>MIMIC_IMAGE_BFL_FALLBACK_DASHSCOPE</code>	BFL failure → DashScope
<code>MIMIC_IMAGE_NVIDIA_FALLBACK_OPENAI</code>	NVIDIA failure → OpenAI
<code>CAROUSEL_COMPOSITE_ENABLED</code>	Sharp composite vs Puppeteer-only

14. Key file map

Area	Files
Flow type constants	<code>src/domain/top-performer-mimic-flow-types.ts</code>
Payload types	<code>src/domain/mimic-payload.ts</code> , <code>src/domain/mimic-carousel-package.ts</code>
Mode classification	<code>src/services/mimic-mode-classifier.ts</code> , <code>src/domain/mimic-text-heavy.ts</code>
Reference resolve	<code>src/services/mimic-reference-resolver.ts</code>
Draft prep	<code>src/services/mimic-draft-prep.ts</code>
LLM copy	<code>src/services/llm-generator.ts</code> , <code>src/services/carousel-mimic-copy-policy.ts</code>
Image generation	<code>src/services/mimic-image-provider.ts</code> , <code>src/services/mimic-image-job.ts</code>
Carousel render	<code>src/services/mimic-carousel-render.ts</code>
Prompts	<code>src/services/mimic-prompt-builder.ts</code>
Project config	<code>src/services/mimic-project-config.ts</code>
Pipeline orchestration	<code>src/services/job-pipeline.ts</code>
HBS template	<code>services/renderer/templates/carousel_mimic_bg.hbs</code>
Docs	<code>docs/MIMIC_IMAGE_FLOWS.md</code> , <code>docs/CREATIVE_INTELLIGENCE.md</code>

15. Invariants and pitfalls

1. `task_id` is the execution key across `content_jobs` , `assets` , `job_drafts` , reviews.
2. `mimic_v1` is render truth; `mimic_carousel_package` is operator/review snapshot — do not conflate with `carousel_package` .
3. **Never** pass full `ideas_json` into mimic LLM prompts.
4. **Image mimic** requires exactly one reference frame — multi-frame references must use carousel mimic.
5. `MIMIC_IMAGE_ENABLED=0` blocks render even if flow types are enabled on the project.
6. **Copy generation** still requires `OPENAI_API_KEY` regardless of image provider.
7. **Re-sign URLs** at render — `refreshMimicPayloadReferenceUrls()` handles expired Supabase signed URLs.
8. **Video slides** in archived carousels are never mimicked.
9. **Mode overrides** on signal packs take precedence over automatic classification.
10. **Do not double-submit** HeyGen or other provider renders — mimic uses its own render path via `job-pipeline.ts` .

16. Glossary

Term	Definition
Top performer	High-performing archived social post used as visual/copy reference
Visual guidelines pack	<code>visual_guidelines_pack_v1</code> on signal pack derived globals
Nemotron	NVIDIA vision model producing <code>aesthetic_analysis_json</code> , <code>text_blocks</code> , <code>mimic_evaluation</code>
Art-only mimic	Image model prompt forbids baking text into pixels
Template bg	Text-heavy deck: extract clean background, overlay fresh copy
Full bleed	Image-led slide: mimic visual plate, overlay text separately
Grounding	Link from signal-pack idea to <code>insights_id</code> of top-performer analysis
Twist brief	Legal constraint: visual pattern only, no verbatim logos/faces

17. Related HTTP endpoints (selection)

- Creative intelligence ingest: `POST /v1/creative-intelligence/:slug/top-performers/ingest`
- Signal pack styling merge: `POST /v1/creative-intelligence/:slug/signal-packs/:id/styling`
- Mimic mode override: signal pack routes in `src/routes/signal-packs.ts`
- Mimic text overlay lab: `src/routes/mimic-text-overlay-lab-routes.ts`
- Review contract paths: covered by `src/routes/review-contract.test.ts`

Cross-linked from `README.md` , `AGENTS.md` , and `docs/CAF_CORE_COMPLETE_GUIDE.md` . For repo-wide doc index see `docs/layers/README.md` .